

Lab Sheet for Compiler

1. Design a DFA to simulate the following machines
 - a. Accepting binary string that start with 0 and ends with 1.
 - b. Accepting the valid variable names in C program
 - c. Accepting the valid gmail ID
 - d. Accepting the prepaid NTC number
2. Design a NFA to simulate the following machine
 - a. Accepting RE $(a + b)^*a$ over $\Sigma = \{a,b\}$
 - b. Accepting RE $(a+b)(a+b)$ over $\Sigma = \{a,b\}$
3. Write a program to remove left recursion from the following grammar
 - a. $S \rightarrow Sab \mid ab \mid a \mid b$ (Grammar 1)
 - b. $A \rightarrow A0 \mid A1 \mid 0$ (Grammar 2)
4. Write a program to create a symbol table for the variables (for data types only)

Example: - Input (int a = 2, float b = 3.5), output will be as

Variable Name	Type	Value	Address
a	integer	2	100
b	real number	3.5	200

5. Write a program to find set of non-terminals, set of terminals, set of productions and starting symbol. Here you have to take the CFG as input in file.

Example:- (For 3a), output will be as

Set of non-terminals = {S}

Set of terminals = {a, b}

Set of productions = { $S \rightarrow Sab$, $S \rightarrow a$, $S \rightarrow b$ }

Starting symbol = {S}

6. Write a program to realize the concept of loop optimization in compiler optimization in following cases, with respect to running time only.
 - a. Code Motion

- i. Example (Consider the following two programs)

Program1

```
for(i=0; i<1000000000; i++) //take the value of counter very large
```

```
{  
    a = b + i ;  
    c = d + e;  
    f = g + h;  
}  
//see the running time lets say t1
```

Program 2(After reducing frequency)

```
c = d + e;  
f = g + h;  
for(i=0; i<1000000000; i++)  
{  
    a = b + i;  
}  
//now see the running time let's say t2  
//then compare the value of t1 and t2
```

b. Loop Jamming